Practice Quiz: Lecture 4 (Part 2) — Line Coding & Pulse Code Modulation

Topic: RZ, Manchester, Block Coding (8B/6T), Digital Representation, PAM, Quantization, PCM Calculations

Student Name: _____	Student ID: _____	Score: _____ / 20

> **Instructions:** Complete all 20 questions. For sampling and bit rate calculations, provide clear formulas and steps in the space provided. Complete solutions are attached at the end.

MCQ **1. What is the major technical drawback of Return-to-Zero (RZ) line encoding compared to NRZ?**

☐ a. It provides no timing transitions for clock synchronization

☐ b. It utilizes only positive voltage levels

☐ c. It requires more signal transitions per bit, thus consuming significantly greater bandwidth

☐ d. It cannot represent binary 0 and binary 1 reliably

☐ e. It produces an unmanageably high DC component in balanced data

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MCQ **2. Which structural feature provides Manchester encoding with its inherent self-synchronization capability?**

☐ a. A mandatory transition at the middle of every bit interval

☐ b. Utilizing four distinct signaling levels

☐ c. Returning to a zero-voltage state between successive bits

☐ d. Varying the duration of individual bit intervals

☐ e. A large baseline DC offset

MCQ **3. Which statement accurately distinguishes Manchester encoding from Differential Manchester encoding?**

☐ a. Manchester uses three voltage levels, while Differential Manchester uses two fixed levels

☐ b. Manchester uses the mid-bit transition for both data and synchronization, whereas Differential Manchester uses the mid-bit transition solely for synchronization and the beginning transition for data

☐ c. Manchester uses the beginning transition for data and clock, whereas Differential Manchester returns to zero mid-bit

☐ d. Differential Manchester never transitions at the middle of a bit interval

☐ e. Manchester has no transitions for bit 0, while Differential Manchester transitions on bit 0 only

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MCQ **4. What is the primary purpose of introducing redundancy in block coding schemes (such as 4B/5B or 8B/10B)?**

☐ a. To convert binary bits directly into continuous analog sinusoids

☐ b. To create longer sequences of uninterrupted zeros in the transmitted code

☐ c. To improve synchronization and increase the reliability of line coding

☐ d. To compress data to reduce the total number of transmitted pulses

☐ e. To remove all transitions from the transmitted waveform

MCQ **5. Which statement correctly describes 8B/6T encoding?**

☐ a. It encodes eight input bits into eight Manchester signal elements

☐ b. It represents each 8-bit data block using six symbols with three voltage levels (ternary)

☐ c. It represents each 6-bit data block using eight distinct voltage levels

☐ d. It converts eight analog samples into six binary data elements

☐ e. It replaces each 8-bit data block with six purely binary pulses

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MCQ **6. What is a key advantage of representing information in digital form rather than analog form?**

☐ a. It eliminates the need for any transmission bandwidth

☐ b. It provides superior tolerance and immunity to noise during signal transmission

☐ c. It completely removes the need for sampling or filtering

☐ d. It allows infinite voltage levels to be detected without error

☐ e. It avoids the need for receiver synchronization

MCQ **7. Which sequence correctly represents the chronological order of basic Pulse Code Modulation (PCM)?**

☐ a. Quantization $\longrightarrow$ Sampling $\longrightarrow$ Encoding

☐ b. Sampling $\longrightarrow$ Quantization $\longrightarrow$ Encoding

☐ c. Encoding $\longrightarrow$ Quantization $\longrightarrow$ Sampling

☐ d. Sampling $\longrightarrow$ Line Coding $\longrightarrow$ Quantization

☐ e. Modulation $\longrightarrow$ Sampling $\longrightarrow$ Encoding

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MCQ **8. Which statement is NOT correct regarding the process of analog-to-digital signal conversion?**

☐ a. Sampling selects signal amplitude values from the analog waveform at discrete periodic time intervals

☐ b. Quantization maps sampled amplitudes into a finite set of predefined discrete levels

☐ c. PAM produces a sequence of pulses whose amplitudes follow the sampled analog signal

☐ d. PAM directly converts each sampled amplitude into its final binary bit sequence

☐ e. Quantized levels are encoded into binary code words for digital transmission

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Calculation **9. An analog voice signal has a highest frequency of $f_{\max} = 6\ \mathrm{kHz}$. Each sample is quantized and represented by $8\ \mathrm{bits}$.**

(a) Calculate the minimum Nyquist sampling rate in Ksamples/s.

(b) Determine the resulting PCM bit rate in Kbps.

Calculation **10. An analog signal has a maximum frequency of $10\ \text{kHz}$ and is quantized into $L = 256$ discrete levels.**

(a) Find the minimum sampling rate in Ksamples/s.

(b) How many bits are required per sample?

(c) What is the resulting PCM bit rate in Kbps?

Calculation **11. An audio channel has a maximum frequency component of $4\ \text{kHz}$. If the signal is sampled at the Nyquist rate and quantized into 16 levels:**

(a) What is the sampling rate?

(b) What is the PCM output bit rate?

Calculation **12. A PCM system samples an analog signal at $8\ \text{Ksamples/s}$ and assigns $7\ \text{bits}$ per sample.**

(a) Calculate the resulting bit rate in kbps.

(b) How many discrete quantization levels L are supported?

Calculation **13. A digital PCM telephone link operates at a bit rate of $128\ \text{kbps}$ with $8\ \text{bits/sample}$. Assuming sampling was performed strictly at the minimum Nyquist rate, what is the maximum frequency of the original analog signal?**

Calculation **14. Compare the minimum bandwidth requirements for transmitting a $10\ \text{Mbps}$ binary stream using:**

(a) NRZ-L (with $r = 1$, $c = 0.5$).

(b) RZ (Return-to-Zero, where a transition occurs mid-bit, yielding $r = 0.5$, $c = 0.5$).

True/False **15. True or False: In Differential Manchester encoding, the presence or absence of a transition at the *beginning* of a bit interval determines the data bit value, while the transition at the middle of the interval serves strictly for synchronization.**

□ a. True

□ b. False

True/False **16. True or False: Increasing the sampling frequency far beyond the Nyquist rate completely eliminates quantization error without needing to increase the number of quantization levels.**

□ a. True

□ b. False

MCQ **17. In the acronym 8B/6T, what does the letter 'T' stand for, and how many distinct signal levels does it signify?**

□ a. Two-phase (2 voltage levels)

□ b. Time-division (8 voltage levels)

□ c. Ternary (3 voltage levels: $+$, 0, $-$)

□ d. Transition-based (4 voltage levels)

□ e. Transmission-code (6 voltage levels)

True/False **18. True or False: Pulse Amplitude Modulation (PAM) produces pulses that are already fully digitized binary bit words suitable directly for digital transmission without quantization.**

□ a. True

□ b. False

MCQ **19. If an analog signal with bandwidth $15\ \text{kHz}$ is sampled at $25\ \text{Ksamples/s}$, what problem occurs at the receiver?**

□ a. Frequency aliasing (distortion), because the sampling rate is less than the Nyquist rate of $30\ \text{Ksamples/s}$

□ b. Baseline wandering, because too many samples are acquired

□ c. Excessive quantization noise, because the bit depth is too high

□ d. Perfect reconstruction without distortion

□ e. DC component inflation

MCQ **20. The mathematical relationship between the number of quantization levels L and the required number of bits per sample n_b is:**

□ a. $n_b = 2^L$

□ b. $n_b = \log_2(L)$

□ c. $L = \log_2(n_b)$

□ d. $n_b = \frac{L}{2}$

□ e. $n_b = 2 \times L$

Answer Key & In-Depth Explanations — Quiz 2

Question 1

Correct Answer: c. It requires more signal transitions per bit, thus consuming significantly greater bandwidth

Explanation: In RZ (Return-to-Zero), the signal must transition to zero halfway through every bit interval. This doubles the signal changes compared to NRZ, requiring twice the bandwidth[cite: 5].

Question 2

Correct Answer: a. A mandatory transition at the middle of every bit interval

Explanation: Manchester encoding guarantees a voltage transition in the exact middle of every bit interval, providing regular clock timing that enables the receiver to synchronize reliably[cite: 5].

Question 3

Correct Answer: b. Manchester uses the mid-bit transition for both data and synchronization, whereas Differential Manchester uses the mid-bit transition solely for synchronization and the beginning transition for data

Explanation: In standard Manchester, the direction of the mid-bit transition signifies the data bit (e.g., low-to-high or high-to-low). In Differential Manchester, the mid-bit transition provides clock timing only, while a transition (or lack thereof) at the start of the bit indicates the bit value[cite: 6].

Question 4

Correct Answer: c. To improve synchronization and increase the reliability of line coding

Explanation: Block coding (such as mapping 4 data bits to 5-bit codewords) introduces redundant bits to prevent long sequences of identical bits, ensuring regular transitions for receiver synchronization and providing error-detection capability[cite: 6].

Question 5

Correct Answer: b. It represents each 8-bit data block using six symbols with three voltage levels (ternary)

Explanation: 8B/6T translates an 8-bit binary pattern into a sequence of 6 ternary signal elements ('T'), each taking one of 3 voltage levels ($+$, 0, $-$)[cite: 6].

Question 6

Correct Answer: b. It provides superior tolerance and immunity to noise during signal transmission

Explanation: Digital signals are discrete; small analog noise perturbations do not alter the interpreted logic level as long as they do not cross the detection threshold, granting digital signals far higher noise tolerance than analog signals[cite: 7].

Question 7

Correct Answer: b. Sampling -> Quantization -> Encoding

Explanation: The standard PCM pipeline converts an analog signal through three successive stages: (1) Sampling (periodic time measurement producing PAM), (2) Quantization (rounding continuous amplitudes into discrete levels), and (3) Encoding (representing quantized levels as binary words)[cite: 7].

Question 8

Correct Answer: d. PAM directly converts each sampled amplitude into its final binary bit sequence

Explanation: PAM (Pulse Amplitude Modulation) produces a series of flat-topped analog pulses proportional in amplitude to the sampled signal. It does NOT generate binary bits; conversion to bits requires subsequent quantization and binary encoding[cite: 7].

Question 9

Correct Answer: (a) 12 Ksamples/s, (b) 96 Kbps

Explanation: The Nyquist sampling theorem requires:

$$f_s = 2 \times f_{\max} = 2 \times 6 \text{ kHz} = 12 \text{ Ksamples/s}[\text{cite: 8}]$$

The PCM bit rate is:

$$\text{Bit Rate} = f_s \times n_b = 12 \text{ Ksamples/s} \times 8 \text{ bits/sample} = 96 \text{ Kbps}[\text{cite: 8}]$$

Question 10

Correct Answer: (a) 20 Ksamples/s, (b) 8 bits, (c) 160 Kbps

Explanation: (a) Minimum sampling rate:

$$f_s = 2 \times f_{\max} = 2 \times 10 \text{ kHz} = 20 \text{ Ksamples/s}[\text{cite: 8}]$$

(b) Bits per sample for $L = 256$:

$$n_b = \log_2(256) = 8 \text{ bits/sample}[\text{cite: 8}]$$

(c) Resulting bit rate:

$$\text{Bit Rate} = f_s \times n_b = 20 \text{ Ksamples/s} \times 8 \text{ bits} = 160 \text{ Kbps}[\text{cite: 8}]$$

Question 11

Correct Answer: (a) 8 Ksamples/s, (b) 32 Kbps

Explanation:

$$f_s = 2 \times 4 \text{ kHz} = 8 \text{ Ksamples/s}[\text{cite: 8}]$$

$$n_b = \log_2(16) = 4 \text{ bits/sample}$$

$$\text{Bit Rate} = 8 \text{ Ksamples/s} \times 4 \text{ bits} = 32 \text{ Kbps}[\text{cite: 8}]$$

Question 12

Correct Answer: (a) 56 kbps, (b) 128 levels

Explanation:

$$\text{Bit Rate} = 8 \text{ Ksamples/s} \times 7 \text{ bits/sample} = 56 \text{ kbps}[\text{cite: 8}]$$

$$L = 2^{n_b} = 2^7 = 128 \text{ quantization levels}[\text{cite: 8}]$$

Question 13

Correct Answer: 8 kHz

Explanation: First find the sampling rate:

$$f_s = \frac{\text{Bit Rate}}{n_b} = \frac{128 \text{ kbps}}{8 \text{ bits}} = 16 \text{ Ksamples/s}[\text{cite: 8}]$$

Since sampling was at the Nyquist rate ($f_s = 2 \times f_{\max}$):

$$f_{\max} = \frac{f_s}{2} = \frac{16 \text{ kHz}}{2} = 8 \text{ kHz}[\text{cite: 8}]$$

Question 14

Correct Answer: (a) 5 MHz, (b) 10 MHz

Explanation: (a) For NRZ:

$$B_{\min} = c \times \frac{N}{r} = 0.5 \times \frac{10 \text{ Mbps}}{1} = 5 \text{ MHz}[\text{cite: 4}]$$

(b) For RZ, two signal states occur per bit interval ($r = 0.5$):

$$B_{\min} = c \times \frac{N}{r} = 0.5 \times \frac{10 \text{ Mbps}}{0.5} = 10 \text{ MHz}[\text{cite: 5}]$$

RZ requires twice the bandwidth of NRZ[cite: 5].

Question 15

Correct Answer: a. True

Explanation: In Differential Manchester, synchronization is maintained by the fixed mid-bit transition, while data is represented differentially by the presence or absence of a transition at the start of the bit[cite: 6].

Question 16

Correct Answer: b. False

Explanation: Sampling above the Nyquist rate prevents spectral aliasing, but quantization error depends on the step size $\Delta = \frac{V_{\max} - V_{\min}}{L}$. To reduce quantization error, one must increase the number of quantization levels L (and consequently the number of bits n_b)[cite: 7, 8].

Question 17

Correct Answer: c. Ternary (3 voltage levels: +, 0, -)

Explanation: In 8B/6T, '8B' denotes 8 binary data bits, and '6T' denotes 6 ternary signal elements having 3 voltage levels[cite: 6].

Question 18

Correct Answer: b. False

Explanation: PAM is an intermediate analog pulse modulation scheme. Its pulse amplitudes take continuous values and are not digital until quantized and converted to binary words[cite: 7].

Question 19

Correct Answer: a. Frequency aliasing (distortion), because the sampling rate is less than the Nyquist rate of 30 Ksamples/s

Explanation: According to the Nyquist theorem, the minimum sampling rate must be $f_s \geq 2 \times f_{\max} = 2 \times 15\ \text{kHz} = 30\ \text{Ksamples/s}$. Sampling at only $25\ \text{Ksamples/s}$ causes overlapping frequency spectral copies (aliasing)[cite: 8].

Question 20

Correct Answer: b. n_b = log2(L)

Explanation: If an analog signal is discretized into L quantization levels, the number of binary bits required to represent each level uniquely is $n_b = \log_2(L)$ (or $L = 2^{n_b}$)[cite: 8].
